Smart Home Quality Assurance - Lab & Test Cases

PART 1 - PROJECT OVERVIEW & SETUP

Project Name: Smart Home QA Web Application
Files Included: index.html, style.css, javascript.js
Testing Credentials:
Email: user@smarthome.com
Password: QApass123

PART 2 - UNDERSTAND THE REQUIREMENTS

- R1 - Navigation: The system should allow users to navigate between SPA sections seamlessly.
- R2 - Add to Cart: The system should allow users to add in-stock smart appliances to the cart.
- R3 - Out of Stock: The system should block users from adding out-of-stock items to the cart.
- R4 - Cart Total: The system should accurately calculate the total price of items in the cart.
- R5 - Empty Cart Checkout: The system should prevent checkout if the cart is empty.
- R6 - Auth Before Checkout: The system must require login before allowing a checkout.
- R7 - Valid Login: The system should allow a registered user to log in.
- R8 - Invalid Login: The system should reject incorrect credentials.
- R9 - Auth Before Service: The system should require login before submitting a QA or Repair request.
- R10 - Submit Service: A logged-in user should successfully submit a QA or Repair request.

PART 3 - TEST SCENARIOS

Scenario ID	Requirement	Test Scenario
TS-01	R1	Verify SPA navigation between Home, About, Shop, and Cart pages.
TS-02	R2	Verify an available item (e.g., Smart Thermostat V2) can be added to the cart.
TS-03	R3	Verify an out-of-stock item

		(e.g., Smart LED Bulb Pack) cannot be added.
TS-04	R4	Verify cart total calculates correctly based on quantity and price.
TS-05	R5	Verify checkout fails when cart is empty.
TS-06	R6	Verify checkout redirects to login if the user is unauthenticated.
TS-07	R7	Verify successful login with valid credentials.
TS-08	R8	Verify login fails with invalid credentials.
TS-09	R9	Verify unauthenticated users cannot submit a QA/Repair form.
TS-10	R10	Verify authenticated users can submit a QA/Repair form.

PART 4 - DETAILED TEST CASES

Test Case 07 - Valid Login
Requirement: R7 - Valid Login
Test Data: Email: user@smarthome.com | Password: QApass123
Steps:
1. Open application
2. Click Login button
3. Enter valid email and password
4. Click Login / Sign Up
Expected Result: Success message appears, button changes to "My Account", redirects to Home.

Test Case 02 - Add to Cart
Requirement: R2 - Add to Cart
Test Data: Smart Thermostat V2
Steps:
1. Navigate to Shop
2. Locate Smart Thermostat V2
3. Click Add to Cart
Expected Result: Cart count increases, success toast notification appears.